

A Modified Set of Constitutive Models for Polymer-Bonded Explosives that Consider Heterogeneity of Initial Cracks and Failure of Damaged Granules

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Abstract: Non-shock ignition of explosions in solids by dynamic loading is a key topic in the field of explosives safety research. Cracks and other damage in polymer-bonded explosives greatly affect non-shock ignition. In this study, we developed a modified set of constitutive models that incorporate heterogeneity of initial cracks and failure of damaged granules, both of which can affect the thermody-

namic response and non-shock ignition of condensed explosives. Moreover, the models were used to predict the distribution of cracks and damage under dynamic compression. The calculated results were in good agreement with experiment, which confirmed the validity of the models.

Keywords: Crack • Failure • Constitutive model • Polymer-bonded explosive • Non-shock ignition

1 Introduction

Damage and non-shock ignition of explosions in solids by dynamic loading present a challenge to production, storage, and transportation of polymer-bonded explosives (PBXs) as they may experience accidental impacts that cause deformation, damage, and fracture that result in energy localization and non-shock ignition. When a condensed-phase explosive is exposed to dynamic loading, its mechanical energy is converted into internal energy and concentrated in several local regions, called hot spots, where a chemical reaction may occur. According to the theory of hot spots, non-shock ignition of explosives under dynamic compression is closely related to mechanisms of crack formation and failure.

In recent decades, much computational and experimental research has been conducted on the mechanisms of crack formation and failure in explosives [1–3]. Dienes [4–6] first established a model of statistical crack mechanics (SCRAM) to account for the opening, shear, growth, and coalescence of an ensemble of cracks. The distribution of cracks in a material is typically assumed to be initially isotropic, and damage anisotropy is captured by tracking the evolution of the mean size of cracks at several orientations. Based on Dienes' SCRAM theory, Addessio *et al.* [7] proposed a simplified, isotropic damage model (ISOSCM) for the dynamic response of brittle materials to nearly isotropic stress. They assumed that during damage propagation, the distribution of cracks remains isotropic and the crack probability-density function is exponential in crack size. Macroscopic crack strains and isotropically damaged surfaces result from averaging the crack strains as well as the instability condition for a single crack under a remote

stress field over all crack orientations. Bennett *et al.* [8] extended ISOSCM to include the viscous effects of binder materials in a plastic bonded explosive, and applied their model (Visco-SCRAM) to study non-shock ignition and mechanical responses of explosives. Zuo *et al.* [9–11] optimized ISOSCM using the dominant crack algorithm (DCA) model. The main feature of the DCA model is a new damage surface that defines the onset of damage evolution. The new damage surface is composed of four surfaces reflecting the strong dependence of critical crack orientation on the state of stress. Wu *et al.* [12, 13] and Wang *et al.* [14] established a micromechanical model combining continuum and discrete damage models in which PBXs are modeled as irregularly shaped energetic particles with discrete binder/particle interfaces. One energetic particle is considered to contain multiple β -cyclotetramethylene tetranitramine (HMX) grains with random orientations that provide for isotropic behavior. A continuum-damage approach was used to model individual HMX particles that contain microcracks. Yang *et al.* [15] developed a viscoelastic-plastic DCA model to describe viscoelastic-plastic deformation and cracking of PBXs in response to mild impacts. The proposed

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model introduces rate-dependent plasticity into the framework, which is more suitable for explosives with relatively high binder content, and damage evolution is calculated using the generalized Griffith instability criterion with dominant crack size rather than crack size averaged over all crack orientations.

In order to directly measure the thermodynamic response of PBXs and calibrate the parameters of constitutive models, a series of impact experiments have been carried out. Williamson *et al.* [16] utilized the Brazilian tests to analyze deformation and damage mechanisms under dynamic loading. It was shown that debonding and cavitation, rather than crystal fracture, were responsible for PBX failure. Si-*viour et al.* [17] measured room temperature stress-strain curves for PBX (EDC 37) at strain rates between 400 s^{-1} and 4000 s^{-1} and showed that compression strength increased linearly with increasing rate over that range. Parab *et al.* [18] used a modified Kolsky bar and high-speed synchrotron X-ray imaging to study in situ deformations and fracture mechanisms in PBXs. Under dynamic compression, most of the cracking in crystals is due to tensile stress generated by diametral compression applied through the contacts between the crystals. Ma *et al.* [19,20] studied the laws of crack initiation and propagation around a cylindrical defect in an explosive simulant using a set of modified split Hopkinson pressure bars equipped with a fluorescence temperature-sensing device. A new phenomenon was observed in which an opening-mode crack emerged from the cylindrical defect earlier than a shearing-mode crack. Furthermore, the rise in temperature in the shearing-mode cracks was higher than that in the opening-mode cracks. These results showed that the defect morphology and strain rate are the main reasons affecting the fracture behavior of PBXs.

Peterson *et al.* [21] carried out a punch-loading experiment using an inert simulant of PBX. The results showed that a dead zone forms in the strain field immediately under the punch in agreement with the slip-line solution of the punch problem. Liu *et al.* [22] investigated shear band development and potential hot-spot initiation in PBXs using a punch-loading experiment at different temperatures. At room temperature, initiation and development of shear bands occurred in three stages and the material suffered a prolonged concentration of shear stress in one area. At high temperatures, initiation and development of shear bands occurred in two stages, with the resulting flow field being close to Hill's solution [23] except for a small area directly underneath the punch.

In the present work, we developed a modified set of constitutive models to describe heterogeneity of the initial cracks and failure of damaged granules, which can affect the thermodynamic response and non-shock ignition of condensed explosives. The model is based on the Visco-SCRAM model and is called the HFV-SCRAM model. The parameters of the HFV-SCRAM model were measured in split Hopkinson pressure bar (SHPB) experiments. Moreover, the

HFV-SCRAM model was used to predict the distribution of cracks and other damage in response to dynamic compression. The Abaqus/Explicit solver combined with the VU-MAT user subroutine was used to calculate two-dimensional stress-strain and damage fields in order to evaluate the HFV-SCRAM model.

2 Model and Method

In this section, we describe the HFV-SCRAM model, which considers heterogeneity of initial cracks and failure of damaged granules.

In the HFV-SCRAM model, the strain rate of the material elements is the sum of those of the viscoelastic and microcrack elements, as given by Eq. (1):

$$\dot{\epsilon}_{ij} = \dot{\epsilon}_{ij}^{ve} + \dot{\epsilon}_{ij}^c \quad (1)$$

where $\dot{\epsilon}_{ij}^{ve}$ is the viscoelastic strain rate and $\dot{\epsilon}_{ij}^c$ is the additional strain rate due to motion between the microcracks.

The viscoelastic strain rate of viscoelastic elements is given by Eq. (2):

$$\dot{\epsilon}_{ij}^{ve} = \frac{\dot{S}_{ij}}{2G} + \frac{S_{ij}}{2\eta} \quad (2)$$

where G is the shear modulus, η is the viscosity coefficient, and S_{ij} is the deviatoric stress. Eq. (2) can be extended to Eq. (3):

$$\dot{S}_{ij} = \sum_{n=1}^N \left(2G^{(n)} \dot{\epsilon}_{ij}^{ve} - \frac{S_{ij}^{(n)}}{\tau^{(n)}} \right) = 2G \dot{\epsilon}_{ij}^{ve} - \sum_{n=1}^N \frac{S_{ij}^{(n)}}{\tau^{(n)}} \quad (3)$$

where $G^{(n)}$ is the shear modulus of the n th Maxwell viscoelastic element, $\tau^{(n)} = \eta^{(n)} / G^{(n)}$ is the relaxation time of the n th Maxwell viscoelastic element, and N is the total number of Maxwell viscoelastic elements.

The additional strain in a microcrack element due to the motion of a microcrack interface is given by Eq. (4):

$$2G\dot{\epsilon}_{ij}^c = \frac{c^3}{c_0^3} S_{ij} \quad (4)$$

The differential form of Eq. (4) is given by Eq. (5):

$$2G\dot{\epsilon}_{ij}^c = 3 \left(\frac{c}{c_0} \right)^2 \frac{\dot{c}}{c_0} S_{ij} + \left(\frac{c}{c_0} \right)^3 \dot{S}_{ij} \quad (5)$$

where c is the mean length of the microcracks, and c_0 is their initial length. The empirical formula for the rate of microcrack growth is given by Eqs. (6), (7), (8), and (9):

$$\dot{c} = v_{\max} \left(\frac{K}{K_1} \right)^m, K < K_1 \quad (6)$$

$$\dot{c} = v_{\max} \left[1 - \left(\frac{K_1}{K} \right)^2 \right], K \geq K_1 \quad (7)$$

$$K = K_0 \sqrt{1 + \frac{2}{m}} \quad (8)$$

$$K_1 = K_0 \sqrt{1 + \frac{2}{m}} \left(1 + \frac{m}{2} \right)^{\frac{1}{m}} \quad (9)$$

where K_0 is the initial fracture toughness of the microcrack elements, K_1 is a critical value related to m , and $v_{\max}$ is the maximum velocity of microcrack propagation. m is the velocity coefficient of microcrack propagation and is obtained by fitting the results of crack growth experiments.

Under low velocity impact loading, the size of a hot spot is usually less than 100 μm . In order to increase the accuracy of the calculations, the size of the finite element mesh should also be less than 100 μm . At this scale, the heterogeneous structure of PBXs needs to be considered. In the traditional model, the initial microcrack length is uniformly distributed in space. However, the initial microcrack length is related to the microstructure of the explosive crystal particles and binder material; this mechanical parameter is non-uniformly distributed in PBXs.

In this work, we utilized a non-uniform distribution of the initial mean length of the microcracks, c_0 , in finite elements to represent heterogeneity of PBX. The Weibull random function was selected to generate non-uniformity in the distribution of c_0 , as given by Eq. (10):

$$c_0 = \theta \cdot x^{\frac{1}{\alpha}} + \mu \quad (10)$$

where the three parameters of the Weibull function are θ (scale), α (shape), and μ (position). $x \in (0, 1)$ is a random number.

PBX is a brittle material. When it is subjected to impact loading, it often breaks or comminutes in local areas. These phenomena change the overall mechanical response of PBXs and lead to generation of hot spots. In the HFV-

SCRAM model, it is assumed that when crack length exceeds twice the characteristic size of the finite element, the finite element is considered to have completely failed. The failed element is regarded as a fluid and can transfer only pressure between the undamaged elements. The HFV-SCRAM model can better describe the behavior of PBXs under compression.

In addition, in order to reflect the influence of confining pressure on fracture toughness, fracture toughness is modified as shown in Eq. (11):

$$K_0 = \beta \cdot P + k_0 \quad (11)$$

where k_0 is fracture toughness without confining pressure, P is confining pressure, and β is the confining pressure strengthening coefficient of fracture toughness.

Two SHPB experiments on HMX-based PBXs were used to calibrate the parameters of the HFV-SCRAM model. The mass fraction of HMX was 95%, binder, 4.3%, insensitive agent, 0.7%, and the density was $1.850 \text{ g} \cdot \text{cm}^{-3}$. Each cylindrical PBX specimen was 5.0 mm long and 10.0 mm in diameter. The mass of each specimen was $726 \pm 5 \text{ mg}$.

The standard SHPB test method was used for stress-strain measurements. The bars were made of 20mm diameter LY12 aluminum alloy. The length of the striker bar, incident bar, transmission bar, and buffer bar were, respectively, 300mm, 2000mm, 1200mm, and 600 mm. An oscilloscope was used to display the strain signals from the incident and transmission bars. The gauge factor, length, and resistance of the strain gauge were, respectively, $2.09 \pm 1.0\%$, 5 mm, $120.4 \Omega \pm 0.4\%$. A schematic illustration of the experimental setup is shown in Figure 1.

3 Results and Conclusion

Two SHPB experiments (Exp. 1 and Exp. 2) were carried out using the same samples. The input conditions of the two SHPB experiments were quite different. The particle velocities in the incident bar in Exp. 1 and Exp. 2 are shown in Figure 2. Wave oscillation resulted from stress wave dispersion in the bars. The loading force of Exp. 1 was much

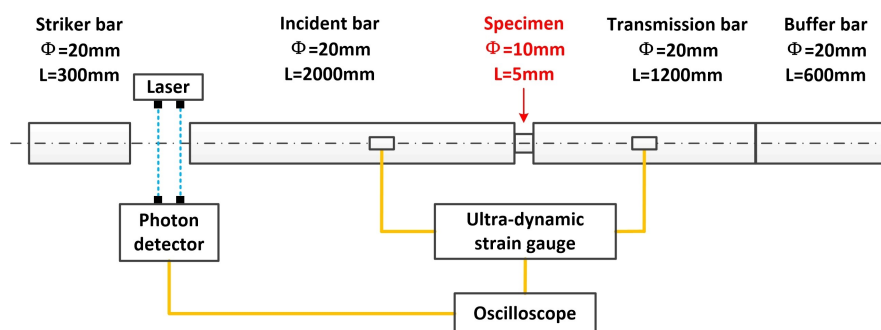


Figure 1. Schematic illustration of the experimental setup.

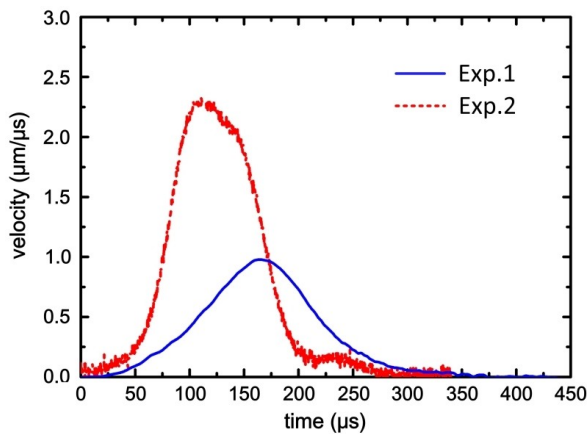


Figure 2. Particle velocity in the incident bar.

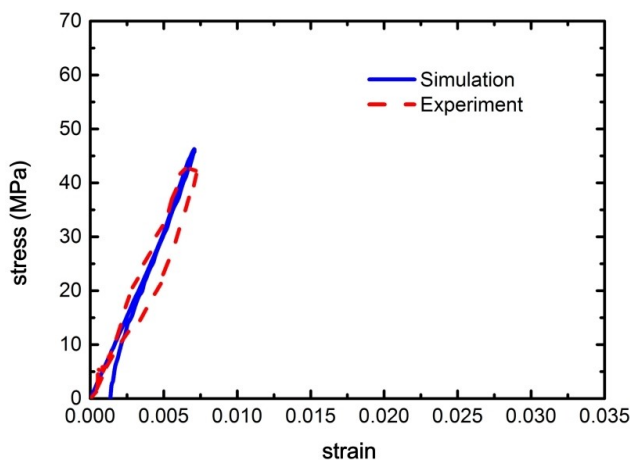


Figure 3. Stress-strain curve of PBX for Exp. 1.

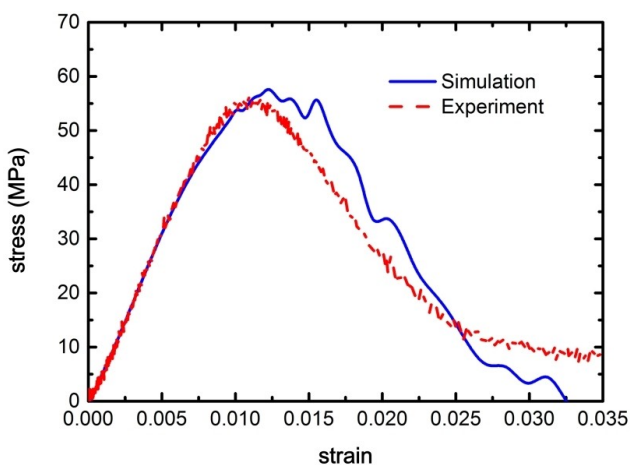


Figure 4. Stress-strain curve of PBX for Exp. 2.

greater than Exp. 2, which was more conducive to calibrating the parameters of the HFV-SCRAM model.

The stress-strain curves of the samples in Exp. 1 and Exp. 2 are shown by the red dashed lines in Figures 3 and 4, respectively. In Exp. 1, the sample was always elastic without obvious damage. In Exp. 2, the sample was in turn elastic, plastic, and stress-softening. The parameters in the HFV-SCRAM model were fitted to the experimental results and are listed in Table 1. The calculated results using the parameters in Table 1 are shown by the blue solid lines in Figures 3 and 4. The stress-strain curves obtained by calculation are in good agreement with the experimental results.

The Abaqus/Explicit solver combined with the VUMAT user subroutine was used to simulate two-dimensional stress and damage fields of PBX in order to study the failure modes. In our finite element code, the grids were established using the medial axis generation algorithm with a 4-node finite element. The constitutive relationships of the specimen were described by the HFV-SCRAM model, and the size of the specimen was 10 mm × 5 mm, consisting of 3600 units of ~80 μm each. The details of the grids are shown in Figure 5. In the simulation, the specimen was at a two-dimensional plane strain condition.

The simulated damage contours in response to SHPB loading are shown in Figures 6 and 7 for Exp. 1 and Exp. 2, respectively, which used simulated initial and boundary conditions consistent with those in the experiments. In Figure 6 and Figure 7, red represents completely damaged areas and blue represents intact areas. In Exp. 1, the maximum loading pressure was about 45 MPa, and the sample had only slight damage at the edge (Figure 8). The HFV-SCRAM model also produced shear damage at the edge of the sample. In Exp. 2, the maximum loading pressure was about 58 MPa, which caused significant damage inside the

Table 1. Critical parameters of the HFV-SCRAM model.

Number	Parameter name	Value
1	ρ_{PBX}	1.85 g/cm ³
2	G_1	3.70 × 10 ⁻³ MPa
3	G_2	0.68 × 10 ⁻³ MPa
4	G_3	2.04 × 10 ⁻³ MPa
5	G_4	3.56 × 10 ⁻³ MPa
6	G_5	2.69 × 10 ⁻³ MPa
7	τ_1	1 × 10 ⁹ μs
8	τ_2	1.37 × 10 ⁻² μs
9	τ_3	1.37 × 10 ⁻¹ μs
10	τ_4	1.37 μs
11	τ_5	0.05 μs
12	c_0	3.0 × 10 ⁻³ cm
13	k_0	1.0 × 10 ⁻⁵ MPa cm ^{1/2}
14	m	10
15	v_{max}	0.11 cm/μs
16	α	0.33
17	θ	3.0 × 10 ⁻³ cm
18	μ	1.0 × 10 ⁻⁹ cm
19	β	0.5

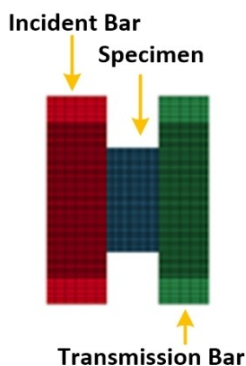


Figure 5. Schematic illustration of the finite element mesh.

experimental sample (Figure 9). The HFV-SCRAM model produced shear damage at 45 degrees to the loading direction, which was consistent with the experimental results. Comparisons between damage contours calculated by numerical simulation and those of the recovered samples showed that the model satisfactorily described material failure in response to compression loading.

Furthermore, the influence of the initial distribution of microcracks on the mechanical response characteristics of materials with the same average microcrack length was investigated by changing the shape parameter α of Weibull random function. The shape parameter α controls the initial distribution of damage in the specimen. The damage here essentially results from the differences in mechanical properties caused by various factors, including the distribution of crystal particle sizes, the proportion of binder, and the distribution of microcrack holes. As α tends towards infinity, the mechanical properties of all parts of the sample become more uniform.

The simulated stress-strain curves were comparable to the experimental values for linear loading of uniform and non-uniform samples, indicating that the initial distribution of microcracks has little effect on initial (elastic) loading (Figure 10). However, there is a significant difference between the two when local damage increases gradually. By contrast with the experimental curve, the curve of the uniform sample does not have a softening section. The loading and unloading curves of the uniform sample closely coincide, indicating that it is largely undamaged with little dissipation during loading and unloading, which approximates elastic behavior. It can be seen from the final damage con-

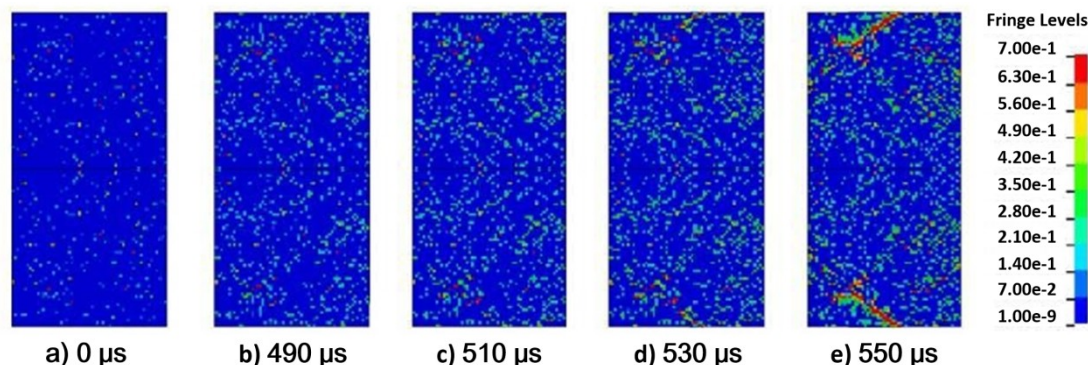


Figure 6. Time sequence of calculated damage contours in response to loading under the conditions of Exp. 1.

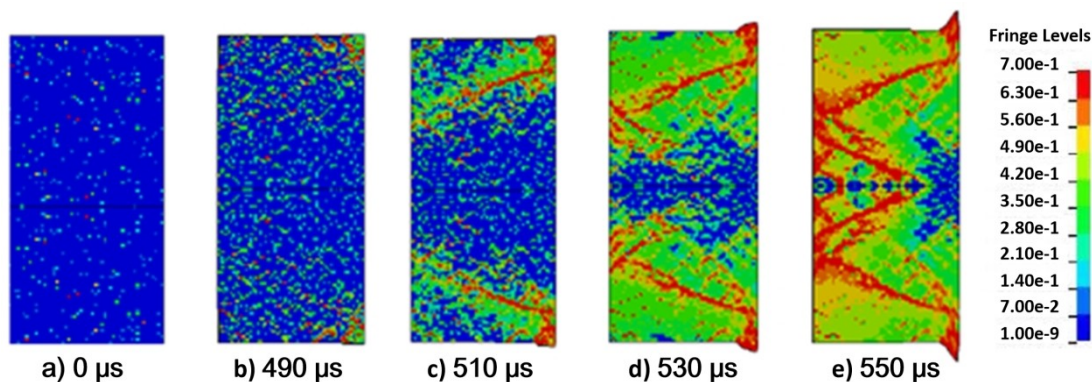


Figure 7. Time sequence of calculated damage contours in response to loading under the conditions of Exp. 2.



Figure 8. Sample recovered from Exp. 1.



Figure 9. Sample recovered from Exp. 2.

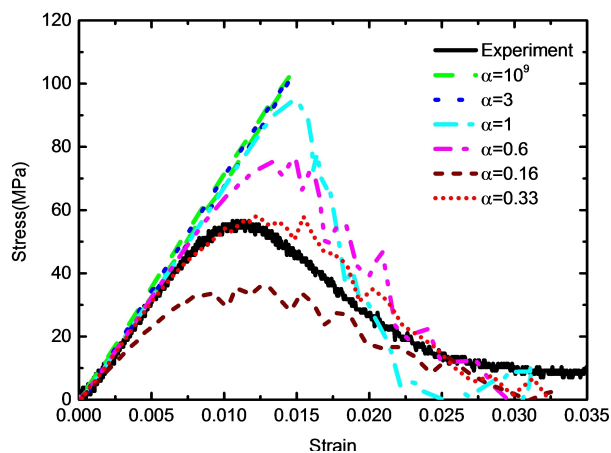


Figure 10. Stress-strain curves for various values of α .

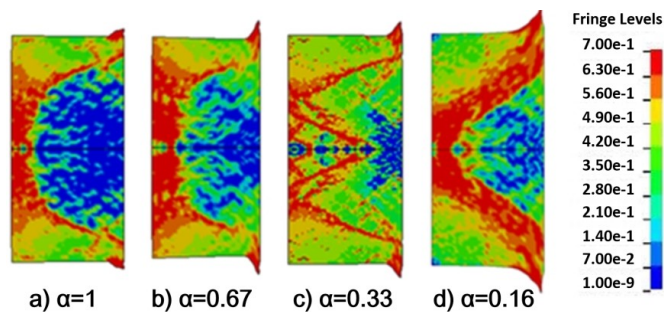


Figure 11. Damage contours for various values of α .

tour (Figure 11) that there is only slight damage at the corners, and the sample remains intact.

With a decrease in the value of the shape parameter α , non-uniformity in the sample becomes more and more apparent. The numerical simulations indicate the sample is more easily damaged, and its compressive strength gradually decreases.

Lower values of uniformity α are associated with reduced sample strength. Moreover, the calculated images of sample failure change with α (Figure 11), which shows that the distribution of internal microstructural damage greatly influences the mechanical properties of the material.

4 Conclusion

A modified set of constitutive models that incorporate heterogeneity of initial cracks and failure of damaged granules was developed to describe the thermodynamic response and non-shock ignition of condensed explosives. The HFV-SCRAM model is based on the Visco-SCRAM model. The parameters of the HFV-SCRAM model were calibrated by split Hopkinson pressure bar (SHPB) experiments. The results showed that lower uniformity (smaller α value) resulted in lower sample strength. Compared with earlier models, the HFV-SCRAM model can better predict the distribution of cracks and damage under dynamic compression.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (No. 11572258 and No. 11902306) and the National Key Laboratory for Shock Wave and Detonation Physics Research Foundation (No. 6142A032020013).

References

- [1] D. B. Hardin, J. J. Rimoli, M. Zhou, Analysis of thermomechanical response of polycrystalline HMX under impact loading through mesoscale simulations, *AIP Adv.* **2014**, *4*, 097136. doi: 10.1063/1.4896699.

- [2] N. Grilli, C. A. Duarte, M. Koslowski, Dynamic fracture and hot-spot modeling in energetic composites, *J. Appl. Phys.* **2018**, 123, 065101. doi: 10.1063/1.5009297.
- [3] R. A. Austin, N. R. Barton, J. E. Reaugh, L. E. Fried, Direct numerical simulation of shear localization and decomposition reactions in shock-loaded HMX crystal, *J. Appl. Phys.* **2015**, 117, 185902. doi: 10.1063/1.4918538.
- [4] J. K. Dienes, On the stability of shear cracks and the calculation of compressive strength, *J. Geophys. Res. [Solid Earth]* **1983**, 88, 1173. doi: 10.1029/JB088iB02p01173.
- [5] J. K. Dienes, A statistical theory of fragmentation processes, *Mech. Mater.* **1985**, 4, 325. doi: 10.1016/0167-6636(85)90030-4.
- [6] J. K. Dienes, Q. H. Zuo, J. D. Kershner, Impact initiation of explosives and propellants via statistical crack mechanics, *J. Mech. Phys. Solids* **2006**, 54, 1237. doi: 10.1016/j.jmps.2005.12.001.
- [7] F. L. Addessio, J. N. Johnson, A constitutive model for the dynamic response of brittle materials, *J. Appl. Phys.* **1990**, 67, 3275. doi: 10.1063/1.346090.
- [8] J. G. Bennett, K. S. Haberman, J. N. Johnson, B. W. Asay, A constitutive model for the non-shock ignition and mechanical response of high explosives, *J. Mech. Phys. Solids* **1998**, 46, 2303. doi: 10.1016/S0022-5096(98)00011-8.
- [9] Q. H. Zuo, F. L. Addessio, J. K. Dienes, M. W. Lewis, A rate-dependent damage model for brittle materials based on the dominant crack, *Int. J. Solids Struct.* **2006**, 43, 3350. doi: 10.1016/j.ijsolstr.2005.06.083.
- [10] Q. H. Zuo, J. K. Dienes, On the stability of penny-shaped cracks with friction: the five types of brittle behavior, *Int. J. Solids Struct.* **2005**, 42, 1309. doi: 10.1016/j.ijsolstr.2004.08.003.
- [11] Q. H. Zuo, J. R. Rice, An implicit algorithm for a rate-dependent ductile failure model, *J. Appl. Phys.* **2008**, 104, 1640. doi: 10.1063/1.3005883.
- [12] Y. Q. Wu, F. L. Huang, A microscopic model for predicting hot-spot ignition of granular energetic crystals in response to drop-weight impacts, *Mech. Mater.* **2011**, 43, 835. doi: 10.1016/j.mechmat.2011.08.004.
- [13] Y. Q. Wu, F. L. Huang, Thermal mechanical anisotropic constitutive model and numerical simulations for shocked β -HMX single crystals, *Eur. J. Mech.* **2012**, 36, 66. doi: 10.1016/j.euro-mechsol.2012.02.011.
- [14] X. J. Wang, Y. Q. Wu, F. L. Huang, Thermal-mechanical-chemical responses of polymer-bonded explosives using a mesoscopic reactive model under impact loading, *J. Hazard. Mater.* **2017**, 321, 256. doi: 10.1016/j.jhazmat.2016.08.061.
- [15] K. Yang, Y. Wu, F. Huang, Numerical simulations of microcrack-related damage and ignition behavior of mild-impacted polymer bonded explosives, *J. Hazard. Mater.* **2018**, 356, 34. doi: 10.1016/j.jhazmat.2018.05.029.
- [16] D. M. Williamson, S. Palmer, W. G. Proud, R. Govier, M. Elert, M. Furnish, W. W. Anderson, W. G. Proud, W. T. Butler, Brazilian disc testing of a uk pbx approaching the glass transition condition, *AIP Conf. Proc.* **2009**, 1195, 494. doi: 10.1063/1.3295182.
- [17] C. R. Siviour, D. M. Williamson, S. G. Grantham, S. Palmer, J. E. Field, Split Hopkinson Bar Measurements of PBXS, *AIP Conf. Proc.* **2004**, 706, 804. doi: 10.1063/1.1780359.
- [18] N. D. Parab, Z. A. Roberts, M. H. Harr, J. O. Mares, A. D. Casey, I. E. Gunduz, M. Hudspeth, B. Claus, T. Sun, K. Fezzaa, S. F. Son, W. W. Chen, High speed X-ray phase contrast imaging of energetic composites under dynamic compression, *Appl. Phys. Lett.* **2016**, 109, 3725. doi: 10.1063/1.4963137.
- [19] X. Ma, X. Li, X. Zheng, K. Li, Q. Hu, J. Li, Crack initiation and potential hot-spot formation around a cylindrical defect under dynamic compression, *J. Appl. Phys.* **2017**, 122, 185104. doi: 10.1063/1.4998679.
- [20] X. Ma, K. Zhang, H. Shang, J. Li, T. Li, H. Fu, X. Zheng, Measuring Crack Growth and Rise in Temperature around a Cylindrical Defect in Explosive Simulants under Low-Pressure and Long-Pulse Loadings, *Propellants Explos. Pyrotech.* **2020**, 45, 1654. doi: 10.1002/prep.202000057.
- [21] P. D. Peterson, K. S. Mortensen, D. J. Idar, B. W. Asay, D. J. Funk, Strain field formation in plastic bonded explosives under compressional punch loading, *J. Mater. Sci.* **2001**, 36, 1395. doi: 10.1023/a:1017572024183.
- [22] Z. W. Liu, H. Y. Zhang, H. M. Xie, K. X. Li, Shear Band Evolution in Polymer Bonded Explosives Subjected to Punch Loading, *Strain* **2016**, 52, 459. doi: 10.1111/str.12186.
- [23] J. M. Hill, W. U. Yong-Hong, The punch problem for shear-index granular materials, *Q. J. Mech. Appl. Math.* **1996**, 1, 81.

Manuscript received: January 31, 2022

Revised manuscript received: July 16, 2022

Version of record online: September 6, 2022